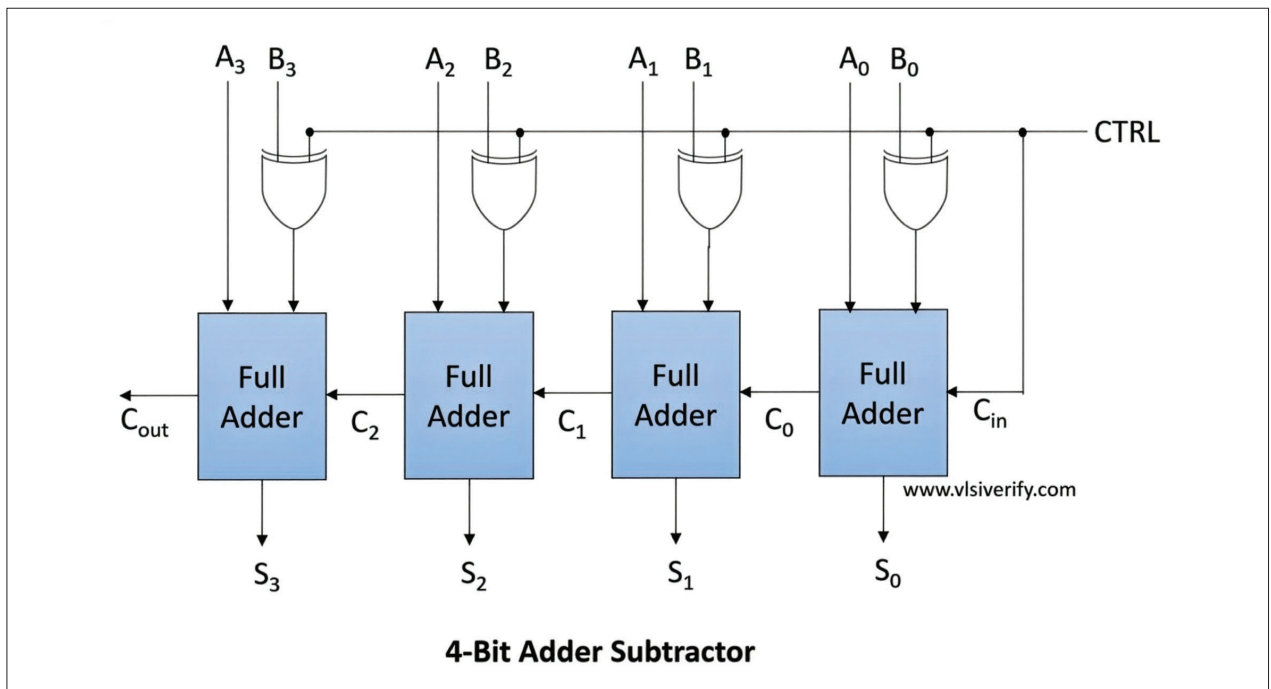




This is combinational logic: output updates based on inputs.
No clock.

Testbench: tb_adder4.v

A testbench is a fake environment that drives your module and checks outputs.

```
`timescale 1ns/1ps

module tb_adder4;
    reg [3:0] a, b;
    reg      cin;
    wire [3:0] sum;
    wire      cout;

    adder4 dut (.a(a), .b(b), .cin(cin),
               .sum(sum), .cout(cout));

    initial begin
        // Wave dump
        $dumpfile("waves/adder4.vcd");
        $dumpvars(0, tb_adder4);

        // Test 1
        a = 4'd3; b = 4'd5; cin = 1'b0;
        #10;

        // Test 2
        a = 4'd15; b = 4'd1; cin = 1'b0;
        #10;
```

```
// Test 3
a = 4'd9; b = 4'd6; cin = 1'b1;
#10;

$finish;
end
endmodule
```

Step-by-step: compile and run (Icarus)

From `ch7_verilog/`:

```
mkdir -p build waves
iverilog -o build/tb_adder4 tb/tb_adder4.v
src/adder4.v
vvp build/tb_adder4
gtkwave waves/adder4.vcd
```

GTKWave supports standard Verilog VCD and is widely used exactly for this “dump then inspect” workflow.

Reading waveforms

Waveforms look intimidating until you know what to look for. Your first checklist:

1. Are inputs changing when I think they are?
2. Does output respond with expected timing?
3. Are there unknowns (X) or high-impedance (Z) where they shouldn't be?
4. Do values glitch unexpectedly? (combinational hazards happen)

GTKWave's user guide emphasizes it's built for debugging large digital systems by analyzing dumped traces, not real-